

Scattering Cross Section of a Small Radome of Arbitrary Shape

ERCUMENT ARVAS, MEMBER, IEEE, AND SAILA PONNAPALLI, STUDENT MEMBER, IEEE

Abstract—A simple moment procedure is used to compute the scattering cross section of a small radome of arbitrary shape. Numerical results are presented for a Von Karman shape radome and a spherical shell. The computed and exact results for the spherical shell are in good agreement.

I. INTRODUCTION

THE PURPOSE OF THIS paper is to present results for the scattering cross section of a radome of arbitrary shape. The maximum linear dimension of the radome is comparable with the wavelength. The equivalence principle is used to replace the radome by equivalent surface electric and magnetic currents radiating in unbounded media. The use of the boundary conditions on the tangential components of the total fields yields a set of coupled integral equations for the surface currents. The combined field formulation is used. The inner and outer surface of the radome are approximated by triangular patches and the method of moment is used to solve the integral equations. Numerical results are presented for a Von Karman shape radome and for a spherical shell. The computed results for the spherical shell are in good agreement with the exact data.

II. SCATTERING FROM A RADOME

The problem of electromagnetic scattering from a dielectric body of arbitrary shape is considered in [1]. Here the method used in [1] is slightly modified to extend it to a dielectric shell of arbitrary shape. The details of the numerical procedure are essentially the same as those used in [1], and hence will be skipped here. Only a brief derivation of the integral equations is given.

The geometry of the problem is represented by Fig. 1(a), where a plane wave is incident on a dielectric shell of permittivity ϵ_2 and permeability $\mu_2 = \mu_0$. The shell is in a homogeneous region characterized by the medium parameters $\epsilon_1 = \epsilon_0$ and $\mu_1 = \mu_0$. In Fig. 1, S_0 and S_i , respectively, denote the outer and the inner surface of the shell and $(\vec{E}^i, \vec{H}^i)$ denote the incident field produced by external sources in the absence of the shell. We are interested in finding the total field $(\vec{E}, \vec{H})$ which is the sum of the incident field and the scattered field $(\vec{E}^s, \vec{H}^s)$. Using the equivalence principle, the original

problem of Fig. 1(a) can be reduced to three simpler but equivalent problems shown in Figs. 1(b)–1(d).

Fig. 1(b) shows the case of the external equivalence. Here we have electric surface current $\vec{J}_0$ and magnetic surface current $\vec{M}_0$ placed on S_0 . These surface currents are radiating in an unbounded medium with parameters (ϵ_1, μ_1) . The same incident field $(\vec{E}^i, \vec{H}^i)$ of Fig. 1(a) is kept here. The total field at any point in the external region (that is the region outside the volume bounded by S_0) in Fig. 1(b) is the same as the total field at the same point of Fig. 1(a). The total field at any point inside the region bounded by S_0 in Fig. 1(b) is zero. In other words, the fields produced by the surface currents, when added to the incident field, sum to the correct total field at external points, and to null field at internal points. Then,

$$\vec{E}_{1 \text{ tan}}(\vec{J}_0, \vec{M}_0) = -\vec{E}_{\text{tan}}^i, \quad \text{on } S_0^- \quad (1)$$

$$\vec{H}_{1 \text{ tan}}(\vec{J}_0, \vec{M}_0) = -\vec{H}_{\text{tan}}^i, \quad \text{on } S_0^- \quad (2)$$

where S_0^- denotes the surface just inside S_0 , and $\vec{E}_1(\vec{J}_0, \vec{M}_0)$ and $\vec{H}_1(\vec{J}_0, \vec{M}_0)$, respectively, denote the electric and magnetic field produced by the surface currents $\vec{J}_0$ and $\vec{M}_0$, when they radiate in the unbounded medium (ϵ_1, μ_1) . The subscript tan denotes the tangential component. The surface currents are related to the total fields as follows:

$$\vec{J}_0 = \vec{n}_0 \times \vec{H}_{\text{out}}^+ \quad (3)$$

$$\vec{M}_0 = \vec{E}_{\text{out}}^+ \times \vec{n}_0. \quad (4)$$

Here $\vec{n}_0$ denotes the unit outward normal vector to S_0 as shown in Fig. 1(a), and $(\vec{E}_{\text{out}}^+, \vec{H}_{\text{out}}^+)$ denote the total fields just outside S_0 in Fig. 1(a).

The field internal to the dielectric region is found using the equivalent problem shown in Fig. 1(c). Here the surface currents $-\vec{J}_0$, $-\vec{J}_i$, $-\vec{M}_0$, and $-\vec{M}_i$ residing on S_0 and S_i , and radiating in the unbounded medium (ϵ_2, μ_2) produce the correct total field $(\vec{E}, \vec{H})$ at any "internal" point, and they produce null field at any external point. Then,

$$\vec{E}_{2 \text{ tan}}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_0^+ \quad (5)$$

$$\vec{E}_{2 \text{ tan}}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_i^- \quad (6)$$

$$\vec{H}_{2 \text{ tan}}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_0^+ \quad (7)$$

$$\vec{H}_{2 \text{ tan}}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_i^- \quad (8)$$

where S_0^+ and S_i^- denote the surfaces just outside S_0 and S_i ,

Manuscript received June 4, 1988; revised October 11, 1988. This work was supported by the Office of Naval Research under Contract N00014-87-K-0366. The computations were conducted using Cornell National Supercomputer Facility.

The authors are with the Department of Electrical and Computer Engineering, Syracuse University, Syracuse, NY 13244.

IEEE Log Number 8927008.

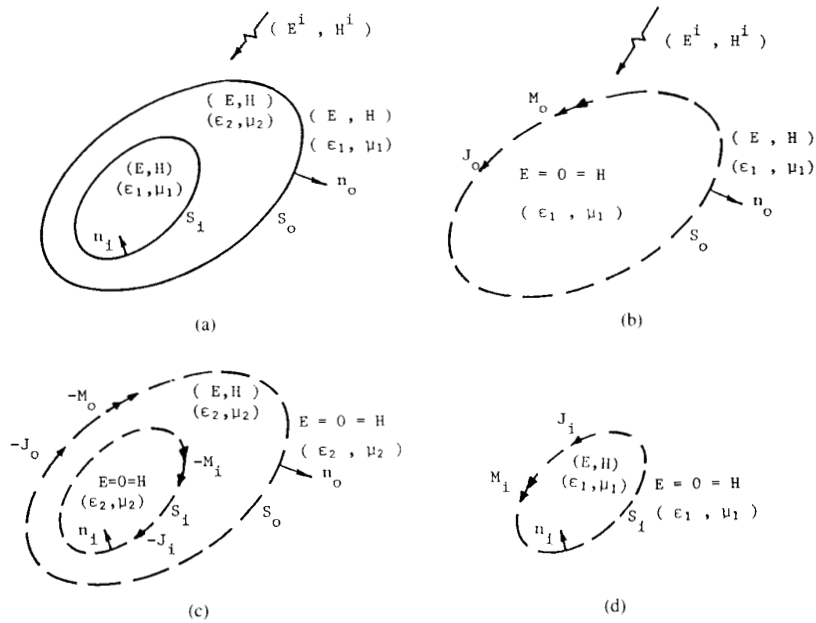


Fig. 1. (a) Geometry of the original problem. (b) The external equivalence. (c) The internal equivalence for the region bounded by S_0 and S_i . (d) The internal equivalence for the region bounded by S_i .

and $\vec{E}_2(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i)$ and $\vec{H}_2(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i)$, respectively, denote the electric and magnetic field produced by the equivalent sources radiating in the unbounded medium (ϵ_2, μ_2) .

The surface currents $\vec{J}_i$ and $\vec{M}_i$ are given by

$$\vec{J}_i = \vec{n}_i \times \vec{H}_{in}^+ \quad (9)$$

$$\vec{M}_i = \vec{E}_{in}^+ \times \vec{n}_i \quad (10)$$

where $(\vec{E}_{in}^+, \vec{H}_{in}^+)$ denote the total fields just inside S_i in Fig. 1(a).

The field in the region bounded by S_i in Fig. 1(a) is found using the equivalent problem shown in Fig. 1(d). Here the surface currents $\vec{J}_i$ and $\vec{M}_i$ residing on S_i and radiating in the unbounded medium (ϵ_1, μ_1) produce the correct total field $(\vec{E}, \vec{H})$ at any internal point, and null field at any point external to S_i . Then

$$\vec{E}_{1 \tan}(\vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_i^+ \quad (11)$$

$$\vec{H}_{1 \tan}(\vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_i^+ \quad (12)$$

Equations (1), (2), (5)–(8), and (11)–(12) represent eight coupled integral equations for the four unknown surface currents $\vec{J}_0, \vec{J}_i, \vec{M}_0, \vec{M}_i$. The combined field formulation reduces these eight equations to four by adding (1) to (5), (6) to (11), (2) to (7) and (8) to (12). Then we have

$$\vec{E}_{1 \tan}(\vec{J}_0, \vec{M}_0) + \vec{E}_{2 \tan}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = -\vec{E}_{\tan}^i, \quad \text{on } S_0 \quad (13)$$

$$\vec{E}_{1 \tan}(\vec{J}_i, \vec{M}_i) + \vec{E}_{2 \tan}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_i \quad (14)$$

$$\vec{H}_{1 \tan}(\vec{J}_0, \vec{M}_0) + \vec{H}_{2 \tan}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = -\vec{H}_{\tan}^i, \quad \text{on } S_0 \quad (15)$$

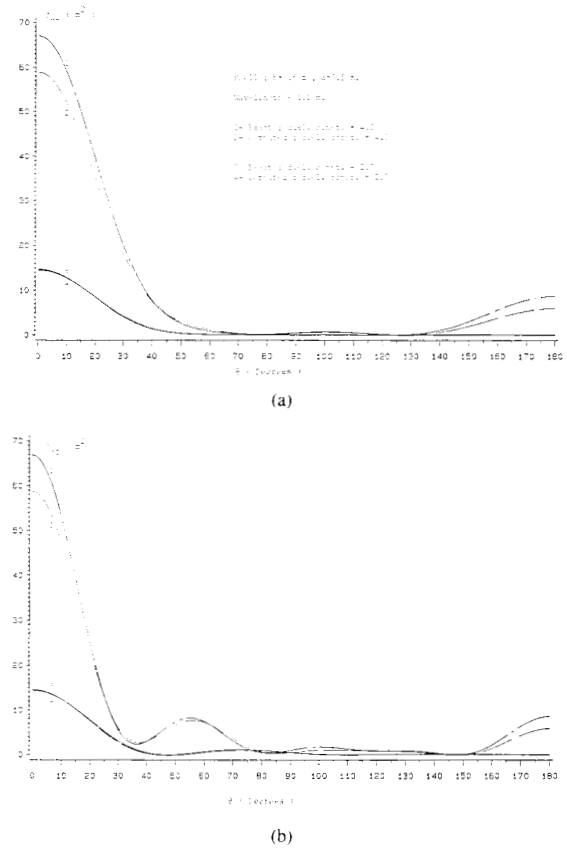


Fig. 2. Comparison of the exact and the computed results for the bistatic scattering cross section of a dielectric spherical shell of inner radius a , and outer radius b . The shell is centered at the origin. (a) σ_{00} (b) σ_{00} .

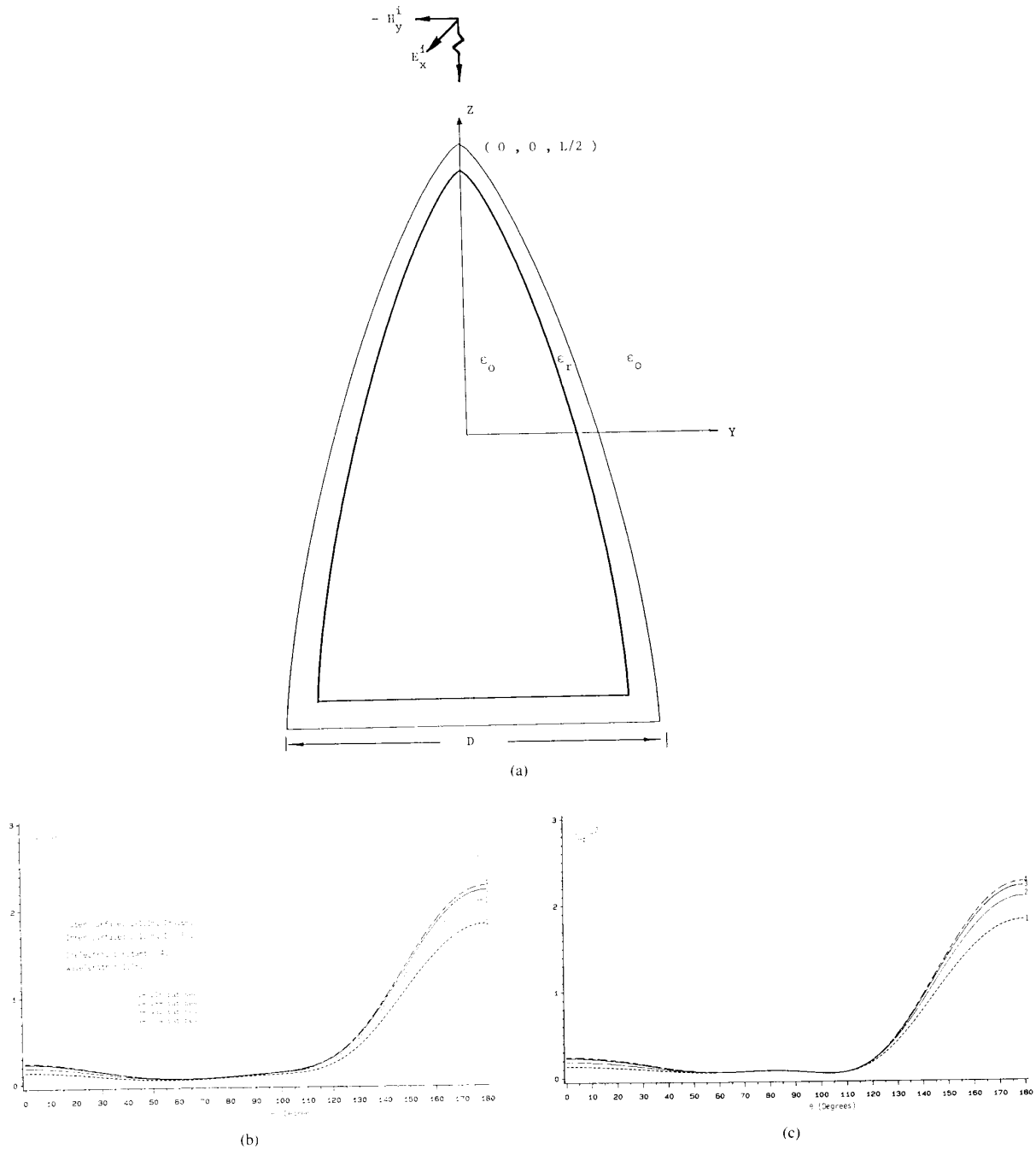


Fig. 3. Plane wave incident on a Von Karman radome. The radome is a body of revolution. Its tip is at $(0, 0, L/2)$. At the back side (the $z = -L/2$ side) the diameter is D . (a) The geometry of the radome. (b) The bistatic scattering cross section σ_{00} . (c) The bistatic scattering cross section σ_{00} .

$$\vec{H}_{1 \tan}(\vec{J}_i, \vec{M}_i) + \vec{H}_{2 \tan}(\vec{J}_0, \vec{M}_0, \vec{J}_i, \vec{M}_i) = 0, \quad \text{on } S_i. \quad (16)$$

These four coupled integral equations can be solved numerically by using the method of moments. The procedure used here is identical to the one used in [1], and hence the details of the numerical procedure are not presented here.

III. SAMPLE RESULTS

Fig. 2 shows the bistatic scattering cross section for a spherical shell of inner radius 0.5λ and outer radius 0.6λ . The shell is centered at the origin and is illuminated by an x-polarized, z-traveling plane wave. Two different dielectric constants are considered, and exact data are also shown for

each case. It is seen that for the lower dielectric constant case the exact and numerical data are in excellent agreement. For the higher dielectric constant case the results are in good agreement except near the forward ($\theta = 0^\circ$) and backward ($\theta = 180^\circ$) directions, where the relative error is still less than 15 percent. For both cases the surface of the shell was approximated by 616 triangular patches. It is seen from these results that for the case of $\epsilon_r = 4.0$, more patches are needed for better conversion because in this case the surface of the shell is more than 15 wavelengths squared.

Fig. 3(a) shows a Von Karman shape radome being illuminated by a plane wave. The surface of a Von Karman shape body is defined by

$$z = \frac{D}{2\sqrt{\pi}} \left\{ \cos^{-1} \left(1 - \frac{2(r+h)}{l} \right) - \frac{1}{2} \cdot \sin \left[2 \cos^{-1} \left(1 - \frac{2(r+h)}{l} \right) \right] \right\}^{1/2}$$

where

$$r = \sqrt{x^2 + y^2}.$$

Figs. 3(b) and 3(c) show the bistatic scattering cross section of the Von Karman radome for various number of patches used to approximate the surfaces. These results are not yet compared with other data.

Further numerical results and details of the solution can be found in [2].

IV. CONCLUSION

A simple moment procedure is used to compute the electromagnetic field scattered by a radome of arbitrary shape. The surface integral equation method is used. Computed results for a spherical shell are in good agreement with the exact data. The method is mainly limited to radomes in the resonance region because as the size of the radome increases, the number of unknowns required for reasonable convergence increases drastically.

REFERENCES

- [1] K. Umashankar, A. Taflov, and S. M. Rao, "Electromagnetic scattering by arbitrary shaped three-dimensional homogeneous lossy dielectric objects," *IEEE Trans. Antennas Propagat.*, vol. AP-34, pp. 758-765, June 1986.
- [2] S. Ponnappalli and E. Arvas, "Electromagnetic scattering from a small radome of arbitrary shape," ECE Dept., Syracuse Univ., Syracuse, NY, Tech. Rep. TR-88-7, June 1988.

Ercument Arvas (M'85), for a photograph and biography please see page 383 of the April 1987 issue of this TRANSACTIONS.



Saira Ponnappalli was born in Berkeley, CA, in 1962. She received the B.S. and M.S. degrees in electrical engineering from Syracuse University, Syracuse, NY, in 1985 and 1988, respectively.

She was employed by the Lockheed Corp. during the summer of 1988 where she was involved with research and development in the k-space iterative approach for solving scattering problems. She is currently pursuing a Ph.D. in electrical engineering at Syracuse University. Her primary area of interest is the application of numerical methods to electro-

magnetic problems.